

2022 FALL

Wednesday 10:10-11:50, 2-C404

Digital Image Processing

GENERAL INTRODUCTION

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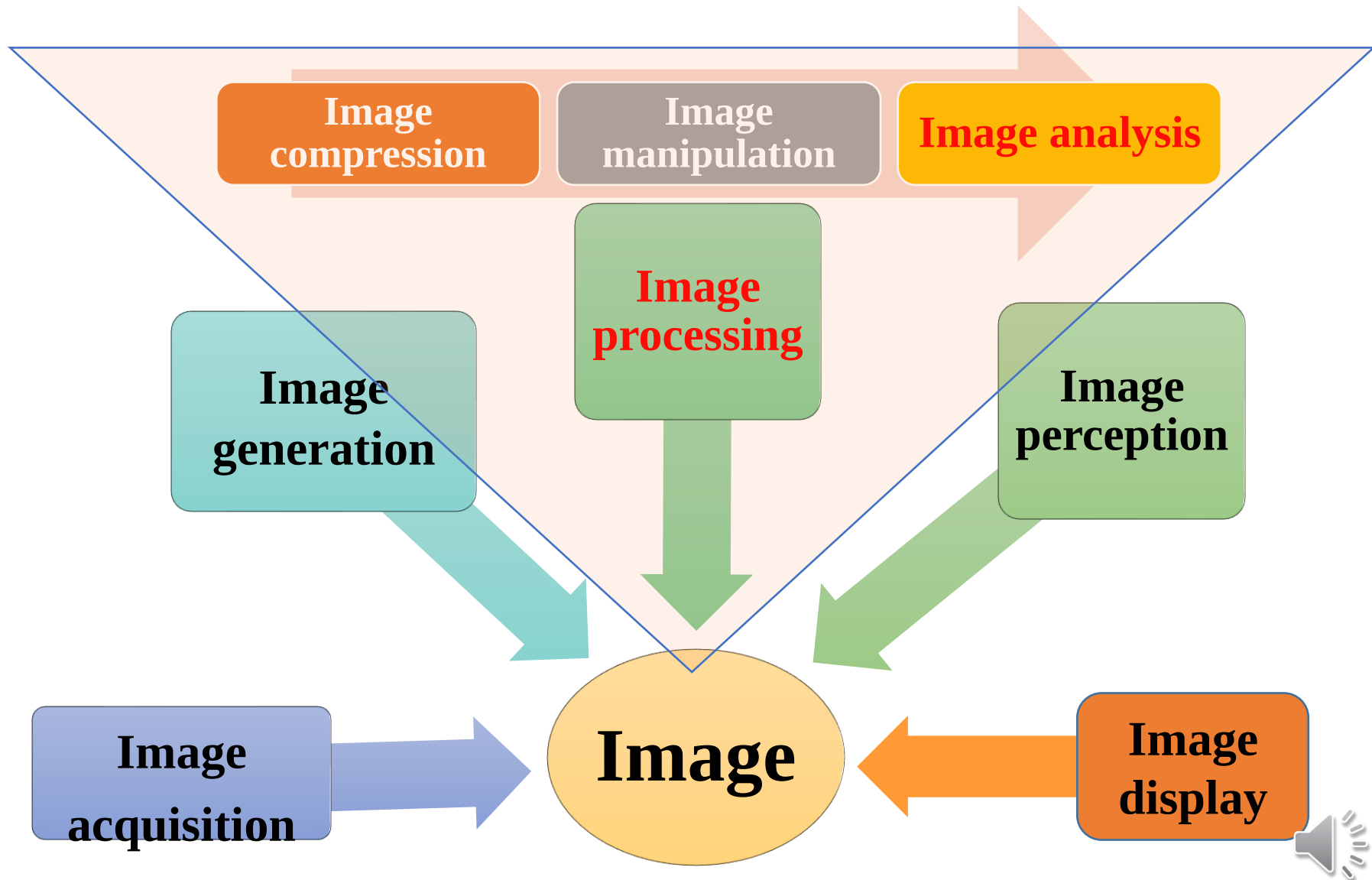


What you can learn from here

- Basic concepts of Digital Image Processing (DIP) and its application.
- Understand basic DIP algorithms.
- Matlab for DIP simulation and solutions.
- Improve your English skills.



DIP where?



Course syllabus

Week	Topic	Hours	Reading
W1	Introduction	2	Ch1
W2	Image Processing Fundamental	2	Ch2
W3	Image Enhancement-Point-based Processing	2	Ch3.1
W4	Image Enhancement - Spatial Filter	2	Ch3.2
W6	Lab1- MATLAB for DIP, Enhancent	4	
W6	Image Enhancement - Frequency Filter		Ch3.3
W7	Edge Detection	2	Ch4
W8	Image Segmentation	2	Ch5



Course syllabus

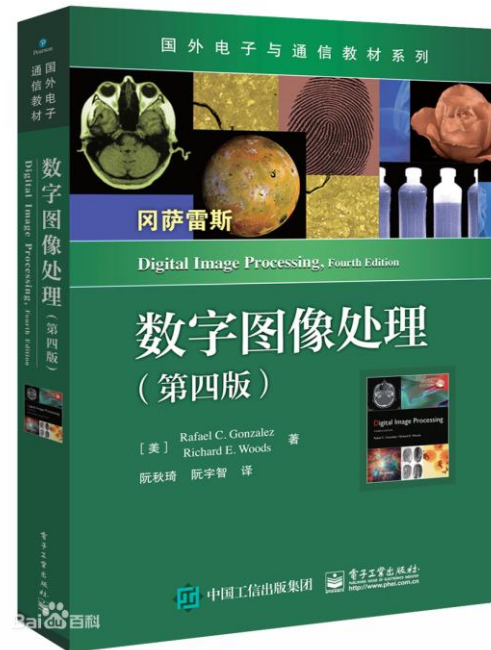
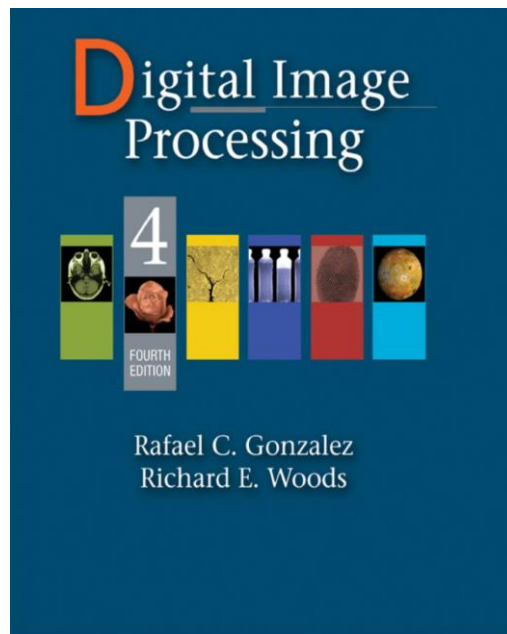
Week	Topic	Hours	Reading
W9	Morphological operation	2	Ch6
W10	Feature extraction and object recognition	2	Ch7/8/9
W11-12	DL/CNN for DIP	4	
W13	Presentation	2	
W14	Presentation	2	
W16	Lab3-Segmenation-final project	4	



Course Material

–Textbook

R. C. Gonzalez, R. E. Woods, *Digital Image Processing*, 4th edition, Prentice Hall, 2018, or similar books



<http://www.imageprocessingplace.com/>

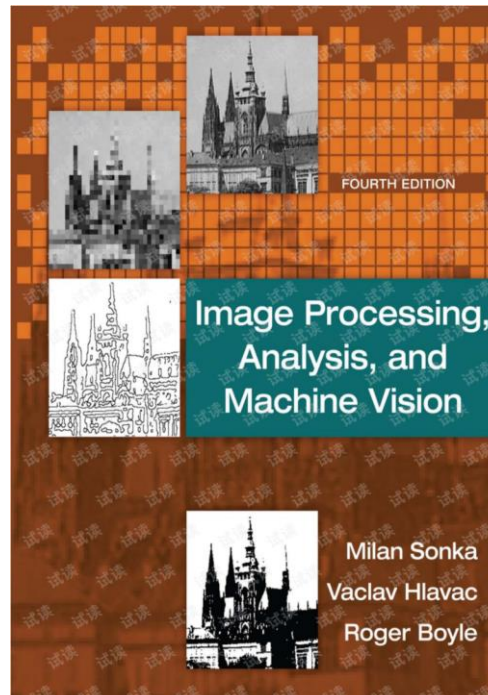


Course Material

–Additional resource

Milan Sonka, Vaclav Hlavac. *Image Processing, Analysis, and Machine Vision*, fourth edition, 2013

Deep learning



–Journals

- IEEE Transactions on Pattern Analysis and Machine Intelligence (T-PAMI)
- IEEE Transactions on Image Processing
- IEEE Transactions on Signal Processing
- IEEE Transactions on Circuits and System for Video Technology
- International Journal on Computer Vision (IJCV)
- Pattern Recognition
- Journal of Visual Communication and Image Representation
- Image and Vision Computing
- Computer Vision and Image Understanding
- Medical Image Analysis
- Graphical Modeling
- Journal of Visual Communication and Image Representation.....

–Conferences

- Vision conferences: CVPR, ICCV, ECCV
- AI conferences: AAAI, IJCAI
- Graphics Conferences



Prerequisite

Mathematics

- ❑ vector & matrix
- ❑ optional, but helpful
- ❑ statistics

Programming language

Python, C++, [Matlab](#), Java, Visual Basic, ...

Optional, but helpful

- ❑ Caffe, Torch, TensorFlow
- ❑ OpenGL
- ❑ IPL(Intel Image Processing Library)
- ❑ OpenCV(Intel Computer Vision Library)...



Grading

Credit Hours

$$28 \text{ (Lecture)} + 8 \text{ (Lab)} + 4 \text{ (Project)} = 40$$

Description	Percentage
Lab	15%
Project presentation	35%
Review report	50%



Labs

- Lab credit hours : $2*4=8$
- When: W6, W16
- Where:

Outlines :

Week	Content	Credit hours
6	Matlab IP TOOLBOX , enhancement	4
16	Image segmentation, final project	4



Evaluation

1. Lab

- Work individually
- Final project at week 16
- 4-5 questions



Evaluation

2. Project presentation

- Work as group (2-4students)
- Work together and prepare PPT
- Run open source code is encouraged
- Each group gives 10 min project presentation
- Project instruction will be given at week 10



Evaluation

3. Final report

- Work individually
- Pick a topic from the list
- Review 20-30 recent papers in DIP field
- Prepare your literature review
- Detailed report instruction will be given at week 10

